

Unifying Special Relativity and Quantum Mechanics

Lecture 8 — Solving the Intertwining Conditions; the (j_1, j_2) Representations

of the Lorentz Group; Decomposition under Rotations and the Klein–Gordon Equation

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Same conventions as Lectures 1–7: everything in the normal text and in the yellow boxes is a faithful transcription of the handwritten notes (yellow boxes correspond to the results that were highlighted in the original, whether in yellow, pink/red, green, or framed); everything inside a blue box like this one is extra material that I added. The figures are redrawn versions of the hand sketches.

Pagination of this scan. The scan continues the triple-starred sequence of Lectures 6–7: its pages carry the circled numbers 25–44, i.e. they are [p. 25***]–[p. 44***]. The first two pages, [p. 25***]–[p. 26***], *repeat the final pages of Lecture 7* (the failure of causality for $[\Phi_+, \Phi_-]_{\pm}$, the combination $\Phi = \kappa\Phi_+ + \eta\Phi_-$, and the inversion of the equation for u^A); since they were already typed there, this document starts from the first genuinely new page, [p. 27***].

Where we left off (recap of Lecture 7). Quantum fields were introduced as the building blocks of causal observables, with the transformation law $U(\Lambda, a)\Phi^A(x)U(\Lambda, a)^\dagger = D(\Lambda^{-1})^A_B \Phi^B(\Lambda x + a)$, and decomposed as $\Phi_{\pm}$ over annihilation/creation operators with coefficient functions $u^A(m, \vec{p}, \sigma) e^{-ip \cdot x}$ and $v^A(m, \vec{p}, \sigma) e^{+ip \cdot x}$. Matching the two ways of transforming the fields produced x -independent consistency conditions; for u^A the lecture ended with the inverted form

$$D(\Lambda)^A_B u^B(m, \vec{p}, \sigma) = \sum_{\sigma'} u^A(m, \vec{p}_\Lambda, \sigma') [D_s(W_L(\Lambda, p))]_{\sigma' \sigma}.$$

This lecture performs the same inversion for v^A , sets up the two-step strategy that solves these conditions, and then builds the toolbox the strategy needs: the classification of the finite-dimensional representations of the Lorentz group by the pair (j_1, j_2) and their decomposition under rotations — ending with the observation that every free field obeys the Klein–Gordon equation.

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1 The final form of the equations for u^A and v^A [p. 27***]

■ Equation for v^A . Starting from

$$\underbrace{D(\Lambda^{-1})^A_B}_{D(\Lambda)^{-1}} v^B(m, \vec{p}_\Lambda, \sigma) = \sum_{\sigma'} v^A(m, \vec{p}, \sigma') [D_s(W_L(\Lambda, p))]_{\sigma\sigma'},$$

we multiply by $D(\Lambda)$ and $D_s(W_L)^{-1}$:

$$\sum_{\sigma'} [D_s(W_L(\Lambda, p))^{-1}]_{\sigma\sigma'} v^A(m, \vec{p}_\Lambda, \sigma') = D(\Lambda)^A_B v^B(m, \vec{p}, \sigma).$$

Using unitarity,

$$(D_s(W_L)^\dagger)_{\sigma\sigma'} = (D_s(W_L)^{T*})_{\sigma\sigma'} = (D_s(W_L))^*_{\sigma'\sigma},$$

$$D(\Lambda)^A_B v^B(m, \vec{p}, \sigma) = \sum_{\sigma'} [D_s(W_L(\Lambda, p))]^*_{\sigma'\sigma} v^A(m, \vec{p}_\Lambda, \sigma')$$

At the end, therefore, we have:

$$D(\Lambda)^A_B u^B(m, \vec{p}, \sigma) = \sum_{\sigma'} [D_s(W_L(\Lambda, p))]_{\sigma'\sigma} u^A(m, \vec{p}_\Lambda, \sigma') \quad (\star)$$

$$D(\Lambda)^A_B v^B(m, \vec{p}, \sigma) = \sum_{\sigma'} [D_s(W_L(\Lambda, p))]^*_{\sigma'\sigma} v^A(m, \vec{p}_\Lambda, \sigma')$$

■ In this form, it is evident that the coefficient functions are “intertwiners” between the unitary irrep of the Poincaré group acting on [p. 28***] one-particle states and the finite-dimensional Lorentz representation carried by the fields.

Added explanation (not in the original notes)

On the index bookkeeping. The unitarity step is the same manipulation used for the u -equation in Lecture 7 ([p. 22***]), now applied in the other direction: it trades the inverse little-group matrix for the complex conjugate, $[D_s(W_L)^{-1}]_{\sigma\sigma'} = [D_s(W_L)]^*_{\sigma'\sigma}$. The final pair (★) agrees with the standard form of the literature (Weinberg, *The Quantum Theory of Fields*, Vol. 1, eqs. (5.1.13)–(5.1.14)): u intertwines with D_s , v with its complex conjugate D_s^* — the residual asymmetry inherited from the fact that a and $a^\dagger$ transform in mutually conjugate representations

(Lecture 6, [p. 8***]). Note the pleasant symmetry of (★): the finite-dimensional D on one side, the unitary little-group D_s on the other, and the coefficient functions in between — “intertwiner” is precisely the representation-theory name for a map that converts one group action into another, as anticipated in the Lecture 7 blue box.

2 Solution strategy: a two-step procedure [p. 28***–30***]

Solution strategy. Two-step procedure:

1) We consider p_{ref} as p , and an element of the little group as Lorentz transformation:

$$\begin{cases} p \rightarrow p_{\text{ref}} \\ \Lambda \rightarrow \Lambda_R(\theta, \hat{n}) \text{ (generic rotation)} \end{cases} \implies \begin{cases} \vec{p}_\Lambda = R(\theta, \hat{n}) \vec{0} = \vec{0} \\ W_L(\Lambda_R(\theta, \hat{n}), p) = \Lambda_R(\theta, \hat{n}) \end{cases}$$

The equation becomes

$$D(\theta, \hat{n})^A{}_B u^B(m, \vec{0}, \sigma) = \sum_{\sigma'} (D_s(\theta, \hat{n}))_{\sigma'\sigma} u^A(m, \vec{0}, \sigma')$$

using the short-hand notation $D(\Lambda_R(\hat{n}, \theta)) \equiv D(\hat{n}, \theta)$ and $D_s(\Lambda_R(\hat{n}, \theta)) \equiv D_s(\hat{n}, \theta)$;

$$D(\theta, \hat{n})^A{}_B v^B(m, \vec{0}, \sigma) = \sum_{\sigma'} (D_s(\theta, \hat{n}))_{\sigma'\sigma}^* v^A(m, \vec{0}, \sigma')$$

Once a choice of “ D ” is made, we can solve these equations to get [p. 29***] $u^A(m, \vec{0}, \sigma)$ and $v^A(m, \vec{0}, \sigma)$.

2) We take again $p = p_{\text{ref}}$, but this time we take as Lorentz transformation $\Lambda = \Lambda_B(\eta, \hat{p})$. There are two consequences:

i) $p_\Lambda = \Lambda p$ becomes

$$p_\Lambda = \Lambda p = \Lambda_B(\eta, \hat{p}) p_{\text{ref}} = p \quad \rightsquigarrow \quad \boxed{p_\Lambda = p}$$

ii) the Wigner rotation reduces to the identity:

$$\rightsquigarrow \quad \boxed{W_L(\Lambda_B(\eta, \hat{p}), p_{\text{ref}}) = \mathbb{1}}$$

From the definition,

$$\begin{aligned} W_L(\Lambda, p) &= \underbrace{\Lambda_B(\hat{p}_\Lambda, |\vec{p}_\Lambda|/M)^{-1}}_{\text{since } p_\Lambda=p, \text{ this is } \Lambda_B(\hat{p}, |\vec{p}|/M)^{-1}} \Lambda \underbrace{\Lambda_B(\hat{p}, |\vec{p}|/M)}_{\text{if } p=p_{\text{ref}}, \text{ this is the identity transformation}} \\ &= \Lambda_B(\hat{p}, |\vec{p}|/M)^{-1} \underbrace{\Lambda_B(\hat{p}, |\vec{p}|/M)}_{\text{this is our choice of } \Lambda} = \mathbb{1}. \end{aligned}$$

The equation thus becomes

$$D(\Lambda)^A{}_B u^B(m, \vec{p}, \sigma) = \sum_{\sigma'} (D_s(W_L(\Lambda, p)))_{\sigma'\sigma} u^A(m, \vec{p}_\Lambda, \sigma') \implies$$

$$D(\Lambda_B(\eta, \hat{p}))^A{}_B u^B(m, \vec{p}_{\text{ref}}, \sigma) = \sum_{\sigma'} \delta_{\sigma'\sigma} u^A(m, \vec{p}, \sigma')$$

[p. 30***] and, in the same way,

$$D(\Lambda)^A{}_B v^B(m, \vec{p}, \sigma) = \sum_{\sigma'} (D_s(W_L(\Lambda, p)))^*_{\sigma'\sigma} v^A(m, \vec{p}_\Lambda, \sigma') \implies$$

$$D(\Lambda_B(\eta, \hat{p}))^A{}_B v^B(m, \vec{p}_{\text{ref}}, \sigma) = \sum_{\sigma'} \delta_{\sigma'\sigma} v^A(m, \vec{p}, \sigma')$$

$$\begin{aligned} u^A(m, \vec{p}, \sigma) &= D(\Lambda_B(\eta, \hat{p}))^A{}_B u^B(m, \vec{0}, \sigma) \\ v^A(m, \vec{p}, \sigma) &= D(\Lambda_B(\eta, \hat{p}))^A{}_B v^B(m, \vec{0}, \sigma) \end{aligned}$$

$\rightsquigarrow$ once we know $u^A(m, \vec{0}, \sigma)$ and $v^A(m, \vec{0}, \sigma)$, we get the coefficient function at generic $\vec{p}$ by boosting.

▼ These equations are analogue to the state transformations

$$|M, \vec{p}, s, \sigma\rangle = U_B(\hat{p}, |\vec{p}|/M) |M, \vec{0}, s, \sigma\rangle$$

but “seen” in the representation of the Lorentz group acting on fields.

What are the possible choices of “ D ”? What is the rationale behind a given choice? Let us consider, to elucidate this point, finite-dimensional representations of the Lorentz group.

Added explanation (not in the original notes)

The logic of the two steps, in one paragraph. The consistency condition (★) must hold for *every* (Λ, p) , which looks like infinitely many equations — but the two clever specializations exhaust its content. *Step 1* ($p = p_{\text{ref}}$, Λ a rotation) probes only the little group: it says the rest-frame coefficients $u^A(m, \vec{0}, \sigma)$ form an invariant map between the spin- s representation D_s (acting on σ) and the restriction of D to rotations (acting on A) — a linear-algebra problem whose solutions are essentially Clebsch–Gordan coefficients, solvable once D is chosen. *Step 2* ($p = p_{\text{ref}}$, Λ the direct boost) probes the coset: since the Wigner rotation trivializes, it doesn’t constrain anything — it *defines* the moving coefficients by boosting the rest-frame ones, in perfect parallel (as the notes emphasize) with the state construction $|M, \vec{p}, s, \sigma\rangle = U_B |M, \vec{0}, s, \sigma\rangle$ of Lecture 5. Any other (Λ, p) is a combination of these two moves, so the general condition is then automatically satisfied (by the cocycle property of W_L). Note that step 2 uses the *spin-polarization* convention (direct boost $L(p) = \Lambda_B(\eta, \hat{p})$): that is why $W_L(\Lambda_B(\eta, \hat{p}), p_{\text{ref}}) = \mathbb{1}$ comes out exactly, with no residual rotation.

3 Finite-dimensional representations of the Lorentz group [p. 31***]

As usual, we start from the algebra. Everything can be understood using angular momentum in QM. From the generators $\vec{J}$ and $\vec{K}$ of the Lorentz group, we define

$$\vec{M} \equiv \frac{\vec{J} + i\vec{K}}{2}, \quad \vec{N} \equiv \frac{\vec{J} - i\vec{K}}{2} \quad \longrightarrow \quad \begin{cases} \vec{J} = \vec{M} + \vec{N} \\ \vec{K} = i(\vec{N} - \vec{M}) \end{cases}$$

with commutators

$$[M^i, M^j] = i\varepsilon_{ijk}M^k, \quad [N^i, N^j] = i\varepsilon_{ijk}N^k, \quad [M^i, N^j] = 0.$$

We “reorganize” the algebra of the Lorentz group in that of two copies of $su(2)$ algebras that do not talk to each other.

↪ Let us use our intuition based on the angular momentum. We diagonalize simultaneously $|\vec{M}|^2, M^3$ and $|\vec{N}|^2, N^3$ by writing

Added explanation (not in the original notes)

Worked check of the three brackets. Using the Lorentz algebra of Lecture 2 ([p. 38]), $[J^i, J^j] = i\varepsilon_{ijk}J^k$, $[J^i, K^j] = i\varepsilon_{ijk}K^k$, $[K^i, K^j] = -i\varepsilon_{ijk}J^k$:

$$[M^i, M^j] = \frac{1}{4} \left(\underbrace{[J^i, J^j]}_{i\varepsilon J^k} + i \underbrace{[J^i, K^j]}_{i\varepsilon K^k} + i \underbrace{[K^i, J^j]}_{-i\varepsilon K^k} - \underbrace{[K^i, K^j]}_{-i\varepsilon J^k} \right) = \frac{i}{4} \varepsilon_{ijk} (2J^k + 2iK^k) = i\varepsilon_{ijk}M^k, \quad \checkmark$$

and analogously for $\vec{N}$ (the cross terms flip sign, giving $J^k - iK^k$), while in $[M^i, N^j]$ all four contributions cancel pairwise. $\checkmark$ The crucial sign making this work is the $-i\varepsilon J$ in $[K, K]$ — the very bracket that distinguishes the Lorentz algebra from the Euclidean one (Lecture 2 blue box). Note that the combinations require the imaginary unit: this innocent-looking i is the seed of the “complexification” remark on [p. 35***], and it also implies that in any finite-dimensional representation where $\vec{M}, \vec{N}$ are Hermitian, $\vec{K} = i(\vec{N} - \vec{M})$ is *anti*-Hermitian — boosts are represented non-unitarily, exactly as stated in the Lecture 7 blue box on $D(\Lambda)$.

4 The (j_1, j_2) classification [p. 32***–34***]

$$|\vec{M}|^2|j_1, m_1\rangle = j_1(j_1 + 1)|j_1, m_1\rangle, \quad M^3|j_1, m_1\rangle = m_1|j_1, m_1\rangle$$

(for fixed j_1 , $-j_1 \leq m_1 \leq j_1$): the set $\{|j_1, m_1\rangle\}_{j_1 \text{ fixed}}$ is the representation space of an irrep of $su(2)$ with index j_1 , $(2j_1 + 1)$ -dimensional.

$$|\vec{N}|^2|j_2, m_2\rangle = j_2(j_2 + 1)|j_2, m_2\rangle, \quad N^3|j_2, m_2\rangle = m_2|j_2, m_2\rangle$$

(for fixed j_2 , $-j_2 \leq m_2 \leq j_2$): the set $\{|j_2, m_2\rangle\}_{j_2 \text{ fixed}}$ is the representation space of an irrep of $su(2)$ with index j_2 , $(2j_2 + 1)$ -dimensional.

Since $[\vec{M}, \vec{N}] = 0$, we can simultaneously diagonalize $|\vec{M}|^2, |\vec{N}|^2, M^3, N^3$ if we take the tensor product of vector spaces. We introduce

$$|j_1, m_1, j_2, m_2\rangle = |j_1, m_1\rangle \otimes |j_2, m_2\rangle$$

(j_1 and j_2 are fixed; these are $(2j_1 + 1) \times (2j_2 + 1)$ -dimensional vectors) with scalar product

$$\langle j_1, m'_1, j_2, m'_2 | j_1, m_1, j_2, m_2 \rangle = \langle j_1, m'_1 | j_1, m_1 \rangle \langle j_2, m'_2 | j_2, m_2 \rangle = \delta_{m_1 m'_1} \delta_{m_2 m'_2},$$

[p. 33***] with eigenvalue equations

$$\begin{aligned} |\vec{M}|^2 |j_1, m_1, j_2, m_2\rangle &= (|\vec{M}|^2 \otimes \mathbb{1}) |j_1, m_1\rangle \otimes |j_2, m_2\rangle = j_1(j_1 + 1) |j_1, m_1, j_2, m_2\rangle, \\ M^3 |j_1, m_1, j_2, m_2\rangle &= m_1 |j_1, m_1, j_2, m_2\rangle, \\ |\vec{N}|^2 |j_1, m_1, j_2, m_2\rangle &= j_2(j_2 + 1) |j_1, m_1, j_2, m_2\rangle, \\ N^3 |j_1, m_1, j_2, m_2\rangle &= m_2 |j_1, m_1, j_2, m_2\rangle, \end{aligned}$$

... and we can also add equations describing the action of $M_{\pm}$ and $N_{\pm}$.

At the conceptual level, therefore, we know how $\vec{M}$ and $\vec{N}$ act on the states $|j_1, m_1, j_2, m_2\rangle$

$\Downarrow$

consequently, we can also reconstruct how $\vec{J}$ and $\vec{K}$ act on $|j_1, m_1, j_2, m_2\rangle$, since $\vec{J} = \vec{M} + \vec{N}$ and $\vec{K} = i(\vec{N} - \vec{M})$

$\Downarrow$

if we now exponentiate a real linear combination, we reconstruct how a generic Lorentz transformation $D(\Lambda) = D(\vec{\eta}, \vec{\theta})$ acts on $|j_1, m_1, j_2, m_2\rangle$.

[p. 34***] ■ Since there is no way of changing j_1 and j_2 once they are fixed, the subspace

$$\left\{ |j_1, m_1, j_2, m_2\rangle \right\}_{\text{fixed } j_1, j_2} \quad \text{with dimension } (2j_1 + 1)(2j_2 + 1)$$

furnishes the representation space of an irrep of the Lorentz group.

The finite-dimensional irreps of the Lorentz group are, therefore, labeled and classified by a pair of indices (j_1, j_2) fixed:

$$D^{(j_1, j_2)}(\Lambda(\omega)), \quad (j_1, j_2) = \begin{cases} (0, 0) \\ (\frac{1}{2}, 0) \\ (0, \frac{1}{2}) \\ \vdots \end{cases}$$

▷ in matrix form, these are $(2j_1 + 1)(2j_2 + 1)$ -dim matrices. In components:

$$\left[D^{(j_1, j_2)}(\Lambda(\omega)) \right]_{m_1 m_2}^{m'_1 m'_2} \equiv \langle j_1, m'_1, j_2, m'_2 | \exp(-i\vec{\theta} \cdot \vec{J} - i\vec{\eta} \cdot \vec{K}) | j_1, m_1, j_2, m_2 \rangle$$

(each basis vector is identified by a pair of indices).

Added explanation (not in the original notes)

A first dictionary of (j_1, j_2) representations. The examples that will populate the rest of the course: $(0, 0)$ is one-dimensional — the *scalar* (this is the trivial representation $D = 1$, zero spin); $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ are the two-component left- and right-handed *Weyl spinors*; their direct sum

$(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ is the four-component *Dirac spinor*; $(\frac{1}{2}, \frac{1}{2})$ is four-dimensional and is exactly the *four-vector* representation $\Lambda^\mu{}_\nu$ of the defining representation; $(1, 0) \oplus (0, 1)$ is the six-dimensional antisymmetric-tensor representation (the one carried by the electromagnetic field strength $F^{\mu\nu}$). The index “ A ” of the field Φ^A of Lecture 7 runs precisely over a basis of one of these spaces. Two remarks: (i) these are irreps of the *algebra* (equivalently, of the cover $SL(2, \mathbb{C})$, see [p. 37***]); the half-integer ones are projective for the Lorentz group itself, exactly as in the $SU(2)$ -vs- $SO(3)$ discussion of Lecture 4; (ii) none of them (except $(0, 0)$) is unitary — consistent with the general theorem recalled in Lecture 7, since here $\vec{K} = i(\vec{N} - \vec{M})$ is manifestly anti-Hermitian.

5 Technical remark: complexification [p. 35***–37***]

$$so(1, 3) = \text{span}_{\mathbb{R}}\{\vec{J}, \vec{K}\}$$

Within this algebra, it is not possible to construct the combinations $\vec{M} = (\vec{J} + i\vec{K})/2$, $\vec{N} = (\vec{J} - i\vec{K})/2$: we need to allow for complex coefficients.

$$so(1, 3)_{\mathbb{C}} = \text{span}_{\mathbb{C}}\{\vec{J}, \vec{K}\}$$

To define $\vec{M}$ and $\vec{N}$, we need to work within the “complexification” of $so(1, 3)$, that is $so(1, 3)_{\mathbb{C}}$.

$$so(1, 3)_{\mathbb{C}} \cong su(2)_{\mathbb{C}} \oplus su(2)_{\mathbb{C}}$$

Since the generators $\vec{M}$ and $\vec{N}$ are defined within a complexified algebra, they generate two commuting subalgebras ($\text{span}_{\mathbb{C}}\{\vec{M}\}$ and $\text{span}_{\mathbb{C}}\{\vec{N}\}$), each isomorphic to $su(2)_{\mathbb{C}}$. [p. 36***]

$$so(1, 3)_{\mathbb{C}} \cong sl(2, \mathbb{C}) \oplus sl(2, \mathbb{C})$$

We exploit the isomorphism $su(2)_{\mathbb{C}} \cong sl(2, \mathbb{C})$.

Flavour of the idea. Consider a basis in $su(2)$ given by $\vec{\sigma}$. Allow for complex combinations, so that a generic combination in $su(2)_{\mathbb{C}}$ is

$$(a_1 + ib_1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + (a_2 + ib_2) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + (a_3 + ib_3) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -u & v \\ w & u \end{pmatrix} :$$

► traceless; ► elements off-diagonal not anymore related; ► this is the Lie algebra $sl(2, \mathbb{C})$, traceless complex 2×2 matrices.

We now exponentiate:

► if we exponentiate a generic complex combination of $\vec{M}$ and $\vec{N}$, we are exploiting the full algebra, and we would get a (complex) representation of the group $SL(2, \mathbb{C}) \otimes SL(2, \mathbb{C})$;

► this is too much. We are only interested in exponentiating a real linear combination of [p. 37***] $\vec{J}$ and $\vec{K}$, with the idea in mind to recover the original Lorentz symmetry. In fact, we obtain something slightly richer: a (complex) representation of the universal covering group $SL(2, \mathbb{C})$ rather than $SO^\uparrow(1, 3)$:

$$\exp(-i\vec{\theta} \cdot \vec{J} - i\vec{\eta} \cdot \vec{K}) \quad (\text{complex}) \text{ irrep of } SL(2, \mathbb{C}).$$

↔ This is a welcome feature because $SL(2, \mathbb{C})$, being the universal covering of $SO^\uparrow(1, 3)$, automatically includes also the projective representations.

↔ The representations are in general complex simply because of the way we constructed them. We used $so(1, 3)_\mathbb{C} \cong sl(2, \mathbb{C}) \oplus sl(2, \mathbb{C})$ and the irreps (j_1, j_2) naturally live in $V = \mathbb{C}^{(2j_1+1) \times (2j_2+1)}$.

Added explanation (not in the original notes)

Why this remark is needed, and what it buys. The manoeuvre $so(1, 3) \rightarrow so(1, 3)_\mathbb{C}$ is the Lie-algebra version of a change of variables that is only allowed over $\mathbb{C}$: the real algebra has no invariant splitting (no real linear combination of $\vec{J}, \vec{K}$ commutes with the rest), but its complexification splits cleanly, and one classifies the representations there — then restricts back to the real combinations $-i\vec{\theta} \cdot \vec{J} - i\vec{\eta} \cdot \vec{K}$ at the end. (This “complexify, classify, restrict” strategy is standard; for compact groups it goes under the name of Weyl’s unitarian trick.) Restricting to real parameters does not undo the classification: the representation spaces and matrices are already fixed by (j_1, j_2) . The chain of isomorphisms also explains cleanly why the universal cover of the Lorentz group is $SL(2, \mathbb{C})$: exponentiating the $(\frac{1}{2}, 0)$ representation, $\vec{J} = \vec{\sigma}/2$, $\vec{K} = -i\vec{\sigma}/2$, gives precisely the traceless exponentials — the 2×2 complex matrices of determinant 1 — and this covers $SO^\uparrow(1, 3)$ two-to-one, in exact analogy with $SU(2) \rightarrow SO(3)$ (Lecture 4, [p. 11*–12*]): the Poincaré-level statement anticipated in the Lecture 4 blue box (Bargmann) is here realized concretely at the Lorentz level.

6 Decomposition under rotations [p. 37***–41***]

We are especially interested in what happens if we restrict our attention to pure rotations. The reason is that, as we discuss, when we [p. 38***] restrict to Lorentz rotations we focus on the spin carried by the field.

We set $\vec{\eta} = 0$, and consider

$$\left[D^{(j_1, j_2)}(\theta, \hat{n}) \right]_{m_1 m_2}^{m'_1 m'_2} = \langle j_1, m'_1, j_2, m'_2 | \exp(-i\theta \hat{n} \cdot \vec{J}) | j_1, m_1, j_2, m_2 \rangle, \quad \vec{J} = \vec{M} + \vec{N}.$$

We can exploit another piece of information from QM: addition of angular momenta. Since $\vec{J} = \vec{M} + \vec{N}$ is the total angular momentum of the two commuting $\vec{M}$ and $\vec{N}$:

► we consider a simultaneous diagonalization of the set $\{|\vec{M}|^2, |\vec{N}|^2, |\vec{J}|^2, J^3\}$ instead of the set $\{|\vec{M}|^2, M^3, |\vec{N}|^2, N^3\}$;

► in essence, this introduces a new basis in the subspace $\{|j_1, m_1, j_2, m_2\rangle\}$ (at j_1, j_2 fixed) given by the vectors

$$\{|j_1, j_2, j, m\rangle\}, \quad |j_1 - j_2| \leq j \leq j_1 + j_2, \quad -j \leq m \leq +j,$$

[p. 39***] defined by the eigensystem

$$\begin{aligned}
|\vec{M}|^2 |j_1, j_2, j, m\rangle &= j_1(j_1 + 1) |j_1, j_2, j, m\rangle, \\
|\vec{N}|^2 |j_1, j_2, j, m\rangle &= j_2(j_2 + 1) |j_1, j_2, j, m\rangle, \\
|\vec{J}|^2 |j_1, j_2, j, m\rangle &= j(j + 1) |j_1, j_2, j, m\rangle, \\
J^3 |j_1, j_2, j, m\rangle &= m |j_1, j_2, j, m\rangle.
\end{aligned}$$

We now ask: what is the operator $D^{(j_1, j_2)}(\theta, \hat{n})$ in this basis? Let us then inspect the form of the matrix

$$\left[D^{(j_1, j_2)}(\theta, \hat{n}) \right]_{jm}^{j'm'} \equiv \langle j_1, j_2, j', m' | \exp(-i\theta \hat{n} \cdot \vec{J}) | j_1, j_2, j, m \rangle.$$

The key aspect is the following. If we consider the matrix element

$$\langle j_1, j_2, j', m' | \exp(-i\theta \hat{n} \cdot \vec{J}) \underbrace{| j_1, j_2, j, m \rangle}_{\substack{\text{this state is still an eigenstate} \\ \text{of } |\vec{J}|^2 \text{ with eigenvalue } j(j+1)}} \stackrel{\textcircled{*}}{=}$$

in fact, we have, using that $[\vec{J}^2, \vec{J}] = 0$: **[p. 40***]**

$$\begin{aligned}
|\vec{J}|^2 \exp(-i\theta \hat{n} \cdot \vec{J}) | j_1, j_2, j, m \rangle &= \exp(-i\theta \hat{n} \cdot \vec{J}) |\vec{J}|^2 | j_1, j_2, j, m \rangle \\
&= \exp(-i\theta \hat{n} \cdot \vec{J}) j(j + 1) | j_1, j_2, j, m \rangle \\
&= j(j + 1) \left[\exp(-i\theta \hat{n} \cdot \vec{J}) | j_1, j_2, j, m \rangle \right].
\end{aligned}$$

$$\stackrel{\textcircled{*}}{=} \delta_{jj'} \langle j_1, j_2, j, m' | e^{-i\theta \hat{n} \cdot \vec{J}} | j_1, j_2, j, m \rangle = \delta_{jj'} [D_j(\theta, \hat{n})]_{m'm}.$$

We find

$$\left[D^{(j_1, j_2)}(\theta, \hat{n}) \right]_{jm}^{j'm'} = \delta_{jj'} [D_j(\theta, \hat{n})]_{m'm}$$

(on the left: $(2j_1 + 1)(2j_2 + 1)$ -dimensional matrix; on the right: $(2j + 1)$ -dimensional matrix). In matrix form:

$$D^{(j_1, j_2)}(\theta, \hat{n}) = \begin{array}{cccc} \boxed{D_{|j_1-j_2|}} & 0 & 0 & 0 \\ 0 & \boxed{\phantom{D_{|j_1-j_2|}}} & 0 & 0 \\ 0 & 0 & \boxed{\phantom{D_{|j_1-j_2|}}} & 0 \\ 0 & 0 & 0 & \boxed{D_{j_1+j_2}} \end{array} \quad \begin{array}{l} \text{each block is} \\ (2j + 1)\text{-dimensional,} \\ \text{orthogonal to} \\ \text{the others} \end{array}$$

[p. 41*]** In compact notation:

$$D^{(j_1, j_2)}(\hat{n}, \theta) \sim \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} D_j(\theta, \hat{n}) \quad \text{decomposition under rotations of a Lorentz irrep.}$$

Comment. This is an important information. Let us go back to the equations for the coefficient functions:

$$D(\theta, \hat{n})^A_B u^B(m, \vec{0}, \sigma) = \sum_{\sigma'} (D_s(\theta, \hat{n}))_{\sigma' \sigma} u^A(m, \vec{0}, \sigma') :$$

on the left, under rotations, D decomposes as $\bigoplus_{j=|j_1-j_2|}^{j_1+j_2} D_j$; on the right, the spin “ s ” is fixed by the kind of particles we want to describe.

If we seek a quantum field capable of describing particles with spin “ s ”, it is necessary for it to transform according to a finite-dimensional representation of the Lorentz group such that its decomposition under rotations includes the representation with $j = s$.

Added explanation (not in the original notes)

Dimension check and first applications of the selection rule. The decomposition is literally the addition-of-angular-momenta theorem of QM applied to $\vec{J} = \vec{M} + \vec{N}$; the dimensions confirm it: $\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (2j_1+1)(2j_2+1)$ (e.g. for $(\frac{1}{2}, \frac{1}{2})$: $1+3=4$ ✓). Applying the boxed selection rule: a spin-0 particle can sit in $(0,0)$ (scalar field, the minimal choice); a spin- $\frac{1}{2}$ particle in $(\frac{1}{2}, 0)$, $(0, \frac{1}{2})$, or their sum (Weyl/Dirac fields); a spin-1 particle needs $j = 1$ in the decomposition, and the minimal Lorentz irreps containing it are $(\frac{1}{2}, \frac{1}{2}) = 0 \oplus 1$ (a vector field A^μ) or $(1,0) \oplus (0,1)$ (an antisymmetric tensor). Note the characteristic mismatch: the vector field carries an unwanted $j = 0$ component alongside the desired $j = 1$ — $2s+1 = 3$ polarizations must be embedded in 4 field components. Removing the excess is precisely the origin of the subsidiary conditions ($\partial_\mu A^\mu = 0$ and, for the massless case, gauge invariance) that make higher-spin field theory subtle; the consistency conditions (★), solved with these D ’s, implement the removal automatically through the explicit form of u^A, v^A . The rule also shows there are infinitely many admissible D ’s for a given s — the choice among them is one of economy and of which interactions one wants to write, a theme presumably developed later.

7 Comment: conjugation [p. 42***]

We saw that in general the irreps of the Lorentz group are complex. It makes sense, therefore, to consider their conjugate.

$\hookrightarrow$ On general grounds, given a representation of a group $D(g)$, its conjugate $D(g)^*$ is still a representation. An important aspect is to understand if $D(g)^*$ gives a genuine new representation or if it is equivalent to $D(g)$, $D(g)^* \sim D(g)$.

In the case of Lorentz, we have

$$\left[D^{(j_1, j_2)}(\Lambda) \right]^* \sim D^{(j_2, j_1)}(\Lambda)$$

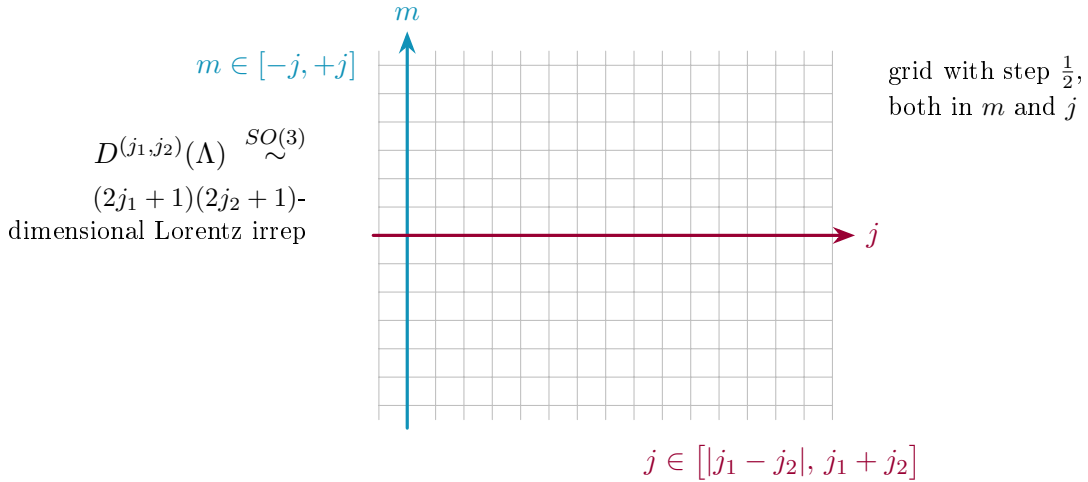
► Lorentz irreps with $j_1 = j_2 = j$ are equivalent to their complex conjugate. *They can be brought in real form with a suitable change of basis.*

Added explanation (not in the original notes)

Why conjugation swaps the labels. Under complex conjugation of the algebra relations, $i \rightarrow -i$, so the combinations $\vec{M} = (\vec{J} + i\vec{K})/2$ and $\vec{N} = (\vec{J} - i\vec{K})/2$ exchange their roles (in the conjugate representation, $-\vec{M}^*$ satisfies the $\vec{N}$ -algebra and vice versa — the $su(2)$ Casimirs get swapped): whatever transformed with weight j_1 under $\vec{M}$ transforms with weight j_1 under $\vec{N}$ in the conjugate, whence $(j_1, j_2)^* \sim (j_2, j_1)$. Consequences worth recording: (i) $(\frac{1}{2}, 0)^* \sim (0, \frac{1}{2})$ — a left-handed Weyl spinor and the conjugate of a right-handed one carry the same representation, the algebraic seed of the particle–antiparticle/chirality pairings of the Standard Model; (ii) self-conjugate representations ($j_1 = j_2$, like the vector $(\frac{1}{2}, \frac{1}{2})$, or parity-symmetric sums like $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ and $(1, 0) \oplus (1, 0)^*$) admit a basis in which the matrices are real — which is why the four-vector field A^μ and the field strength $F^{\mu\nu}$ can be taken real-valued, while a single Weyl field is irreducibly complex. This dovetails with the conjugate representations already met at the operator level: a vs $a^\dagger$ (Lecture 6) and the u - vs v -equations in (★).

8 The (j, m) grid schematic [p. 43***]

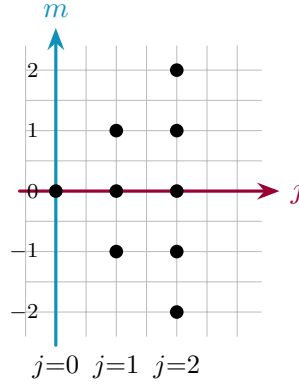
We introduce the following schematic.



We organize the $(2j_1 + 1)(2j_2 + 1)$ components of (j_1, j_2) into a (j, m) grid that makes evident the decomposition under rotations.

Example:

$$D^{(1,1)}(\Lambda) \sim^{SO(3)} \bigoplus_{j=0}^2 D_j$$



Added explanation (not in the original notes)

Reading the grid. Each column j hosts the $2j + 1$ points $m = -j, \dots, +j$ of one rotation block D_j ; the total number of dots equals the dimension of the Lorentz irrep. The example $(1, 1)$: $9 = 1 + 3 + 5$ components splitting as $j = 0 \oplus 1 \oplus 2$ — this is the representation carried by a symmetric traceless rank-2 tensor field $h^{\mu\nu}$ minus its trace parts, the natural habitat of a spin-2 particle (the graviton): the selection rule of [p. 41***] is satisfied ($j = 2$ appears), at the cost of the extra $j = 0, 1$ components to be disposed of by constraints/gauge symmetry — the same pattern noted for the vector field, one notch up in spin.

9 All free fields satisfy the Klein–Gordon equation [p. 44***]

Comment. Going back to

$$\Phi^A(x) = \sum_{\sigma} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\kappa u^A(m, \vec{p}, \sigma) e^{-ip \cdot x} a(m, \vec{p}, \sigma) + \eta v^A(m, \vec{p}, \sigma) e^{ip \cdot x} a^\dagger(m, \vec{p}, \sigma) \right],$$

if we apply the box operator:

$$\begin{aligned} \square \Phi^A(x) &= \partial_\mu \partial^\mu \Phi^A(x) = \sum_{\sigma} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\kappa u^A(m, \vec{p}, \sigma) \underbrace{(-ip_\mu)(-ip^\mu)}_{-p^2 = -M^2} e^{-ip \cdot x} a(m, \vec{p}, \sigma) \right. \\ &\quad \left. + \eta v^A(m, \vec{p}, \sigma) \underbrace{(ip_\mu)(ip^\mu)}_{-p^2 = -M^2} e^{ip \cdot x} a^\dagger(m, \vec{p}, \sigma) \right] \\ &= -M^2 \Phi^A(x). \end{aligned}$$

That implies

$$(\square + M^2) \Phi^A(x) = 0 \quad \text{KLEIN–GORDON EQUATION}$$

$\rightsquigarrow$ All free fields, irrespectively on their spin, must satisfy the Klein–Gordon equation.

Added explanation (not in the original notes)

What the Klein–Gordon equation is (and is not). The computation shows that KG is nothing but the mass-shell condition $p^2 = M^2$ of the Wigner irrep (Lecture 3) read in position

space through the plane-wave factors $e^{\mp i p \cdot x}$ with $p^0 = E_p$ on-shell: it is a *kinematic* statement, valid component by component for every spin, encoding “the quanta have mass M ”. It is *not*, in general, the full field equation: for spin 0 it exhausts the free dynamics, but for higher spins the intertwining conditions (★) impose more — first-order equations (Dirac) or transversality/constraint equations (Proca, Maxwell) that also remove the unwanted $j \neq s$ components identified in the [p. 41***] blue box, and which *imply* KG rather than being implied by it. Historically the equation was proposed (by Schrödinger, Klein, Gordon, and others) as a relativistic replacement of the Schrödinger equation for a wave function, and abandoned because of negative-energy and probability-interpretation troubles; in the present framework those troubles evaporate: Φ^A is an *operator*, its two frequency components multiply a and $a^\dagger$ (both with positive energy E_p), and the state space is the Fock space of Lecture 6. The derivation above is a neat capstone: the whole group-theoretic chain — irreps $\rightarrow$ creation operators $\rightarrow$ fields — lands back on the most familiar equation of relativistic wave mechanics, now correctly interpreted.

(End of Lecture 8 — the notes continue, presumably, with the explicit solution of the intertwining conditions for the simplest choices of $D^{(j_1, j_2)}$, starting from the scalar field.)